

Winding Sectors and the Four-Displacement Restricted-Permutation Problem

A direct Tribonacci–Lucas enumeration, an inverse transseries,
and corrections to OEIS A000382, A000496, and A008305

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Abstract

For $n \geq 4$, let Φ_n be the number of permutations σ of $\mathbb{Z}/n\mathbb{Z}$ for which every clockwise displacement $\sigma(i) - i \pmod{n}$ belongs to $\{0, 1, 2, 3\}$. Equivalently, $\Phi_n = \text{per}(I + P + P^2 + P^3)$ for the cyclic shift matrix P . This is the fourth column of OEIS A008305 and the tail of A004306. Current OEIS entries also identify it with A000496 and with four times A000382, but those sequences first disagree at $n = 8$.

We give a short direct proof of the exact formula

$$\Phi_n = 2(L_n + 1), \quad L_0 = 3, \quad L_1 = 1, \quad L_2 = 3, \quad L_n = L_{n-1} + L_{n-2} + L_{n-3}.$$

(The formula itself is already printed in OEIS A004306, as “Equals $2^*(A001644(n)+1)$, $n \geq 3$ ”; what is new here is its bijective proof.) The proof partitions every allowed permutation by its integer winding number. A constant-cut lemma shows that winding-one permutations are precisely cyclic tilings by monomers, dimers, and trimers; reflection-complementation identifies the winding-two sector with the winding-one sector, while the two outer sectors are singletons. The method extends to the outer winding sectors for all displacement widths.

Consequences include the corrected recurrence and rational generating function, an exact Binet formula, exponentially sharp asymptotics, a nearest-integer formula, and an all-orders logarithmically oscillatory inverse transseries. We also prove the recurrence currently marked “conjectured” in A000382, but for a different reason: it is an identity for the sequence produced by Mendelsohn’s displayed 1961 formula, not for the restricted- permutation count. The exact correction is

$$\sum_{n \geq 4} \left(\frac{\Phi_n}{4} - A000382(n) \right) z^{n-4} = \frac{z^4}{1 - 2z + z^4}.$$

An independent subset-dynamic-programming verifier is included. This is a research draft; the historical and novelty claims should be independently checked before an OEIS or journal submission.

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1 The problem, the discrepancy, and the result

1.1 The cyclic restricted-permutation count

Write the elements of $\mathbb{Z}/n\mathbb{Z}$ as $0, 1, \dots, n-1$. For a permutation $\sigma: \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$, let d_i be the unique integer in $\{0, 1, \dots, n-1\}$ satisfying

$$\sigma(i) = i + d_i \pmod{n}.$$

For $n \geq 4$ define

$$\Phi_n := \#\{\sigma \in S(\mathbb{Z}/n\mathbb{Z}) : d_i \in \{0, 1, 2, 3\} \text{ for every } i\}. \quad (1.1)$$

If P is the permutation matrix for the clockwise cyclic shift, then

$$\Phi_n = \text{per}(I + P + P^2 + P^3). \quad (1.2)$$

This is Mendelsohn's $\phi(n, 4)$ in the notation of his 1961 paper [1]; it is also $T(n, 4)$ in OEIS A008305 [8].

The first values obtained from the definition are

$$24, 44, 80, 144, 264, 484, 888, 1632, 3000, 5516, 10144, \dots \quad (n = 4, 5, 6, \dots). \quad (1.3)$$

They are recorded in OEIS A004306, whose current entry gives the recurrence $a(n) = 2a(n-1) - a(n-4)$ and a Tribonacci formula [7]. *[Write note, batch 73O2, 1 October 2026.]* Its cross-reference line already records the equivalent form $a(n) = 2(A001644(n) + 1)$ for $n > 3$, and a tribonacci formula is credited there to N. L. Skirrow (February 2026); Theorem 1.1 supplies a bijective proof of that form. The formula $\Phi_n = 2(A001644(n) + 1)$ is therefore not new; the contributions of this report are its proof by winding sectors, the general outer-sector theorem, and the diagnosis of A000382 and A000496 (Section 1.4). The purpose of this report is not to rediscover that recorded recurrence by sequence fitting. It is to give a transparent bijection explaining it, to separate three historically entangled sequences, and to derive several consequences that do not appear in the checked entries.

1.2 Three sequences that agree only through n=7

As of October 1, 2026, OEIS A000382 begins

$$6, 11, 20, 36, 65, 119, 218, 400, 735, \dots$$

with offset 4, and says that it is the fourth column of A008305 divided by 4; it also marks

$$a(n) = a(n-1) + a(n-2) + a(n-3) - 2$$

as conjectural [5]. A000496 begins

$$1, 1, 2, 6, 24, 44, 80, 144, 260, 476, 872, \dots$$

and is likewise named “Restricted permutations” [6]. However, A008305 itself gives 264, 484, 888 in the relevant column at $n = 8, 9, 10$, and A004306 gives the same values [8, 7].

The divergence is summarized below. “Equation (4)” means the sequence obtained by evaluating Mendelsohn's displayed formula, while “printed table” means the values printed on page 32 of the same paper. *[Write note, batch 73O2: the two Mendelsohn rows are as read from a scan of the 1961 paper; the scan is not in the repository, and those values were not re-checked when the report was placed. The other two rows were recomputed from the definition and from the OEIS snapshots.]*

n	4	5	6	7	8	9	10
actual Φ_n	24	44	80	144	264	484	888
Mendelsohn equation (4)	24	44	80	144	260	476	872
Mendelsohn printed table	24	44	80	144	256	472	872
$4 A000382(n) = A000496(n)$	24	44	80	144	260	476	872

Thus the source formula, the source table, and the permanent are three distinct objects. The first discrepancy between the current A000382 comment and the actual fourth column occurs at

$$n = 8 : \quad A000382(8) = 65, \quad \frac{T(8, 4)}{4} = \frac{264}{4} = 66.$$

The issue is not a matter of offset convention.

1.3 Main theorem

Let L_n denote the Tribonacci–Lucas sequence

$$L_0 = 3, \quad L_1 = 1, \quad L_2 = 3, \quad L_n = L_{n-1} + L_{n-2} + L_{n-3}. \quad (1.4)$$

This is OEIS A001644 [9].

Theorem 1.1 (Exact four-displacement enumeration). *For every $n \geq 4$,*

$$\boxed{\Phi_n = 2(L_n + 1)}. \quad (1.5)$$

More precisely, if the allowed permutations are sorted by winding number $q \in \{0, 1, 2, 3\}$, then the four sector sizes are

$$(1, L_n, L_n, 1). \quad (1.6)$$

The proof occupies Sections 2–4. It uses only a conservation law and a cyclic-tiling bijection.

1.4 What is new here, and what is not claimed

The recurrence and generating function for the correct count are already present in A004306, and related transfer-matrix formulae occur in the classical literature on permanents of cyclic $(0, 1)$ matrices [2, 3]. *[Write note, batch 7302: so is the formula $\Phi_n = 2(A001644(n) + 1)$ of Theorem 1.1, in the cross-reference line of A004306, together with Skirrow’s tribonacci formula; item (i) below is a proof of a recorded formula, not a new formula.]* The contributions of this report are:

- (i) a direct winding-sector proof of the exact formula;
- (ii) a general theorem identifying the outer winding sectors for every fixed displacement width;
- (iii) a source-level diagnosis of the A000382/A000496 discrepancy;
- (iv) a proof of the A000382 recurrence for the formula-defined, rather than permutation-defined, sequence;
- (v) the exact correction kernel and its asymptotics;

(vi) exact and all-orders asymptotic formulae, including an inverse log-periodic transseries.

No claim is made that no equivalent bijection has ever appeared in the wider literature. The literature search was targeted rather than exhaustive. The OEIS statements quoted here are snapshots and may be corrected after the date of this report.

2 Winding sectors and a constant-cut law

It is useful to treat a slightly more general problem. Fix integers $n \geq r \geq 2$, and define

$$\Phi_{n,r} := \#\{\sigma \in S(\mathbb{Z}/n\mathbb{Z}) : \sigma(i) - i \pmod{n} \in \{0, 1, \dots, r-1\}\}. \quad (2.1)$$

For every allowed permutation choose the representatives $d_i \in \{0, 1, \dots, r-1\}$.

Lemma 2.1 (Integral winding number). *For every allowed permutation,*

$$q(\sigma) := \frac{1}{n} \sum_{i=0}^{n-1} d_i \quad (2.2)$$

is an integer in $\{0, 1, \dots, r-1\}$.

Proof. Because σ is a permutation,

$$\sum_i \sigma(i) \equiv \sum_i i \pmod{n}.$$

Hence $\sum_i d_i \equiv 0 \pmod{n}$. The bounds $0 \leq d_i \leq r-1$ give $0 \leq q \leq r-1$. $\square$

Draw the vertices $0, 1, \dots, n-1$ clockwise on a circle. Represent the arrow $i \mapsto i + d_i$ by the clockwise arc that traverses exactly d_i elementary cuts. Let cut k be the gap from k to $k+1$, and let c_k be the number of nontrivial arcs crossing that cut.

Lemma 2.2 (Constant-cut law). *Every cut is crossed by exactly $q(\sigma)$ arcs:*

$$c_0 = c_1 = \dots = c_{n-1} = q(\sigma). \quad (2.3)$$

Proof. Moving from cut $k-1$ to cut k , the crossing count gains one if a nonloop arc starts at k and loses one if a nonloop arc ends at k . There is one outgoing and one incoming arrow at every vertex. Moreover, the outgoing arrow at k is a loop if and only if the incoming arrow at k is that same loop: if $\sigma(k) = k$, no other arrow can end at k , and conversely. Therefore the number of nonloop starts at k equals the number of nonloop ends at k , so $c_k - c_{k-1} = 0$.

The sum of the n cut counts is the total number of elementary cuts traversed by all arcs, namely $\sum_i d_i = qn$. Since the cut counts are constant, each is q . $\square$

This elementary observation is the structural core of the argument. It turns a modular displacement condition into a covering problem on a circle.

2.1 Reflection-complementation

Define an operation on allowed permutations by

$$(\mathcal{C}_r\sigma)(x) := r - 1 - \sigma(-x) \pmod{n}. \quad (2.4)$$

It is the composition of reflection in the domain, the permutation σ , and an affine reflection in the codomain.

Lemma 2.3 (Sector symmetry). *The map $\mathcal{C}_r$ is an involution, and it sends winding sector q to winding sector $r - 1 - q$. In particular, the winding-sector polynomial*

$$W_{n,r}(u) := \sum_{\sigma} u^{q(\sigma)} \quad (2.5)$$

is palindromic of degree $r - 1$.

Proof. If d_{-x} is the displacement of σ at $-x$, then

$$(\mathcal{C}_r\sigma)(x) - x \equiv r - 1 - \sigma(-x) - x \equiv r - 1 - d_{-x} \pmod{n}.$$

The displayed integer already lies in $\{0, \dots, r - 1\}$. Summing over x changes qn into $(r - 1)n - qn$. A second application of $\mathcal{C}_r$ returns σ . $\square$

The extreme sectors are immediate.

Lemma 2.4 (Outer singleton sectors). *The winding-0 sector consists only of the identity, and the winding- $(r - 1)$ sector consists only of the shift $i \mapsto i + r - 1$.*

Proof. An average of numbers in $[0, r - 1]$ equals 0 only when every number is 0, and it equals $r - 1$ only when every number is $r - 1$. $\square$

3 Winding one is a cyclic-tiling problem

Let $k = r - 1$. A cyclic tiling of a labeled n -cycle by tiles of lengths $1, 2, \dots, k$ is a partition of its elementary cuts into clockwise intervals of those lengths. A tile has a distinguished initial vertex, so the labels and orientation are retained.

Theorem 3.1 (Outer-sector tiling theorem). *For $n \geq r$, the winding-1 permutations counted by $\Phi_{n,r}$ are in bijection with cyclic tilings of a labeled n -cycle by tiles of lengths $1, 2, \dots, r - 1$.*

Under the bijection, a tile of length j becomes the displacement block

$$j \underbrace{00 \cdots 0}_{j-1 \text{ zeros}}. \quad (3.1)$$

The winding- $(r - 2)$ sector has the same size by Lemma 2.3.

Proof. By Lemma 2.2, every cut is covered by exactly one arc in a winding-1 permutation. Consequently, the cut intervals traversed by its nonloop arcs are pairwise disjoint and cover the cycle. If an arc starts at i and has length j , then no nonloop arc can start at an internal vertex $i + 1, \dots, i + j - 1$,

because its first cut would already be covered. Those internal vertices are therefore loops, giving the block in (3.1). The nonloop arcs thus define a cyclic tiling.

Conversely, replace each tile of length j by an arc from its initial vertex to its terminal vertex, and put loops on the $j - 1$ internal vertices. Every internal vertex is its own target, while every tile initial vertex is the target of the preceding tile's arc. Thus every vertex has exactly one preimage and the construction is a permutation. Each cut is crossed once, so its winding number is 1. The two constructions are inverse. $\square$

For $r = 4$, the three tile blocks are particularly simple:

$$1, \quad 20, \quad 300. \quad (3.2)$$

For example, the cyclic composition $8 = 3 + 2 + 1 + 2$ gives the displacement word

$$300\,20\,1\,20 = 30020120.$$

3.1 Counting the cyclic tilings

Let $F_m^{(k)}$ be the number of linear tilings of length m with tile lengths $1, \dots, k$. Set $F_0^{(k)} = 1$ and $F_m^{(k)} = 0$ for $m < 0$. Then

$$F_m^{(k)} = \sum_{j=1}^k F_{m-j}^{(k)}, \quad \sum_{m \geq 0} F_m^{(k)} z^m = \frac{1}{1 - z - z^2 - \dots - z^k}. \quad (3.3)$$

Proposition 3.2 (Cyclic-tiling count). *Let $C_{n,k}$ be the number of cyclic tilings of a labeled n -cycle with tile lengths $1, \dots, k$. Then, for $n \geq 1$,*

$$C_{n,k} = \sum_{j=1}^k j F_{n-j}^{(k)}. \quad (3.4)$$

It satisfies the same order- k recurrence as $F_m^{(k)}$ once the initial range has passed. Equivalently, if $\lambda_1, \dots, \lambda_k$ are the roots of

$$x^k - x^{k-1} - \dots - x - 1 = 0, \quad (3.5)$$

then

$$C_{n,k} = \lambda_1^n + \dots + \lambda_k^n. \quad (3.6)$$

Proof. Focus on the unique tile containing vertex 0. If that tile has length j , then vertex 0 can occupy any of its j labeled positions. Removing it leaves a line of length $n - j$, which can be tiled in $F_{n-j}^{(k)}$ ways. This proves (3.4).

The power sums in (3.6) obey the recurrence associated with (3.5). Newton's identities give the same initial values as (3.4); alternatively, both sides are the trace of the n th power of the standard composition automaton. Hence they agree. $\square$

The dominant root λ_k of (3.5) is real, simple, and lies in $(1, 2)$. Multiplication by $x - 1$ shows that it is the nontrivial root in $(1, 2)$ of

$$x^{k+1} - 2x^k + 1 = 0. \quad (3.7)$$

Thus the outer winding sector already explains why generalized Fibonacci roots appear in confined-displacement problems.

4 Complete solution for four displacements

Set $r = 4$ and $k = 3$. The possible winding numbers are 0, 1, 2, 3. There are no unaccounted-for inner sectors: winding 0 and 3 are singletons, winding 1 is the cyclic-tiling sector, and winding 2 is its reflected complement.

Write simply $F_m = F_m^{(3)}$. Thus

$$F_0 = 1, \quad F_1 = 1, \quad F_2 = 2, \quad F_m = F_{m-1} + F_{m-2} + F_{m-3}. \quad (4.1)$$

By Proposition 3.2,

$$C_n = F_{n-1} + 2F_{n-2} + 3F_{n-3}. \quad (4.2)$$

The initial values are

$$C_0 = 3, \quad C_1 = 1, \quad C_2 = 3,$$

and therefore $C_n = L_n$ from (1.4). This proves Theorem 1.1.

A compact refinement is the full winding polynomial.

Corollary 4.1 (Sector polynomial). *For $n \geq 4$,*

$$W_{n,4}(u) = 1 + L_n u + L_n u^2 + u^3. \quad (4.3)$$

4.1 Recurrences

Substituting $L_n = (\Phi_n - 2)/2$ into the Tribonacci recurrence gives the affine order-three recurrence

$$\boxed{\Phi_n = \Phi_{n-1} + \Phi_{n-2} + \Phi_{n-3} - 4} \quad (n \geq 7), \quad (4.4)$$

with $\Phi_4 = 24, \Phi_5 = 44, \Phi_6 = 80$. Subtracting two consecutive instances gives the homogeneous order-four recurrence

$$\boxed{\Phi_n = 2\Phi_{n-1} - \Phi_{n-4}} \quad (n \geq 8). \quad (4.5)$$

The Tribonacci–Lucas numbers are odd: the first three are odd, and each later term is a sum of three odd terms. Hence the theorem also proves

$$4 \mid \Phi_n \quad (n \geq 4). \quad (4.6)$$

Define the corrected quarter sequence

$$R_n := \frac{\Phi_n}{4} = \frac{L_n + 1}{2}. \quad (4.7)$$

It begins

$$6, 11, 20, 36, 66, 121, 222, 408, 750, 1379, 2536, 4664, \dots \quad (4.8)$$

and satisfies

$$R_n = R_{n-1} + R_{n-2} + R_{n-3} - 1 \quad (n \geq 7). \quad (4.9)$$

4.2 Generating functions

Let the tail variable start at $n = 4$. From (4.5),

$$\begin{aligned} \sum_{m \geq 0} \Phi_{m+4} z^m &= \frac{24 - 4z - 8z^2 - 16z^3}{1 - 2z + z^4} \\ &= \frac{4(6 - z - 2z^2 - 4z^3)}{(1 - z)(1 - z - z^2 - z^3)}. \end{aligned} \quad (4.10)$$

Dividing by four gives

$$\boxed{\sum_{m \geq 0} R_{m+4} z^m = \frac{6 - z - 2z^2 - 4z^3}{1 - 2z + z^4}.} \quad (4.11)$$

If one includes the exceptional small- n values 1, 1, 2, 6 appearing in A004306, then

$$\sum_{n \geq 0} A004306(n) z^n = \frac{1 - z + 2z^3 + 13z^4 - 3z^5 - 6z^6 - 10z^7}{1 - 2z + z^4}, \quad (4.12)$$

which agrees with the current A004306 entry [7].

5 The Mendelsohn formula sequence and A000382

This section separates a valid recurrence identity from an invalid combinatorial identification.

5.1 The formula-defined sequence

Mendelsohn defines the third-order Fibonacci sequence in his indexing as

$$1, 2, 4, 7, 13, 24, 44, 81, 149, \dots$$

With our $F_0 = 1, F_1 = 1, F_2 = 2, \dots$, his displayed equation (4), divided by four, gives for $n \geq 5$

$$M_n = F_{n-1} + F_{n-3} + F_{n-4} + 1, \quad (5.1)$$

with the separately correct value $M_4 = 6$. The sequence M_n is exactly the current A000382 tail and $4M_n$ is the current A000496 tail.

There is a harmless typographical issue in the preceding paragraph's display in some text extractions of the 1961 scan; the sequence meant is the one printed in the paper and used in (5.1). The numerical evaluation is unambiguous: $M_8 = 65, M_9 = 119, M_{10} = 218$.

Theorem 5.1 (Resolution of the A000382 recurrence). *The formula-defined sequence M_n satisfies*

$$\boxed{M_n = M_{n-1} + M_{n-2} + M_{n-3} - 2} \quad (n \geq 8). \quad (5.2)$$

Its shifted generating function is

$$\boxed{\sum_{m \geq 0} M_{m+4} z^m = \frac{6 - z - 2z^2 - 4z^3 - z^4}{1 - 2z + z^4}.} \quad (5.3)$$

Thus the recurrence marked conjectural in A000382 is a theorem for the sequence defined by (5.1). It is not the recurrence of the actual quarter-count R_n .

Proof. The sequence $M_n - 1$ is a fixed linear combination of three shifts of F_n , so it obeys the homogeneous Tribonacci recurrence whenever all shifted instances of (5.1) are valid. Therefore

$$M_n - 1 = (M_{n-1} - 1) + (M_{n-2} - 1) + (M_{n-3} - 1),$$

which is (5.2). The initial tail 6, 11, 20, 36, 65 then gives the numerator in (5.3) after multiplication by $1 - 2z + z^4$. $\square$

The homogeneous recurrence $M_n = 2M_{n-1} - M_{n-4}$ begins only at $n = 9$; it fails at $n = 8$ because the numerator in (5.3) contains a nonzero z^4 term. This one-step distinction is another useful check on indexing.

5.2 The exact correction kernel

Subtracting (5.3) from (4.11) gives the central provenance identity.

Theorem 5.2 (Exact correction to A000382 and A000496). *For the current A000382 tail M_n and the true quarter-count R_n ,*

$$\boxed{\sum_{n \geq 4} (R_n - M_n) z^{n-4} = \frac{z^4}{1 - 2z + z^4} = \frac{z^4}{(1 - z)(1 - z - z^2 - z^3)}}. \quad (5.4)$$

Consequently,

$$R_n - M_n = \begin{cases} 0, & 4 \leq n \leq 7, \\ \sum_{j=0}^{n-8} F_j, & n \geq 8. \end{cases} \quad (5.5)$$

For A000496, the correction is four times this amount.

The first nonzero corrections to A000382 are therefore

$$1, 2, 4, 8, 15, 28, 52, 96, 177, \dots \quad (n = 8, 9, 10, \dots). \quad (5.6)$$

They are not a finite transcription error: they grow exponentially at the same Tribonacci rate as the two compared sequences.

A useful closed form for the cumulative composition numbers is

$$\sum_{j=0}^m F_j = \frac{F_{m+2} + F_m - 1}{2}. \quad (5.7)$$

It follows either by induction or by multiplying generating functions.

5.3 What the 1961 source actually contains

Mendelsohn's paper gives the correct definition of $\phi(n, r)$ and presents its evaluation via permanents [1]. For $r = 4$, however, its displayed equation (4) produces 260, 476, 872 at $n = 8, 9, 10$, while its printed table gives 256, 472, 872 (as read from the scan; see the note in Section 1). The permanent gives 264, 484, 888. The agreement through $n = 7$ explains how the formula-defined sequence could remain attached to the intended combinatorial description.

Plouffe's 1992/2009 collection explicitly describes its formulae as computer-generated approximations and conjectures from the sequence database available at the time [4]. Its rational generating function correctly captures the inherited A000382 terms; it cannot repair a prior error in those terms without returning to the underlying definition.

6 Exact roots, asymptotics, and nearest integers

Let α be the real root of

$$x^3 - x^2 - x - 1 = 0, \quad (6.1)$$

and let β, γ be the two nonreal roots. Numerically,

$$\alpha = 1.8392867552141611325518525646532866004\dots, \quad (6.2)$$

$$\beta = \rho e^{i\theta}, \quad \gamma = \rho e^{-i\theta},$$

$$\rho = \alpha^{-1/2} = 0.7373527057603276752017596505081211233\dots, \quad (6.3)$$

$$\theta = 2.1762335454918703984542598307236195688\dots \quad (6.4)$$

Indeed $\alpha\beta\gamma = 1$, so $|\beta| = |\gamma| = \alpha^{-1/2}$, and

$$\cos \theta = \frac{(1 - \alpha)\sqrt{\alpha}}{2}. \quad (6.5)$$

Theorem 6.1 (Exact Binet form). *For every $n \geq 4$,*

$$\Phi_n = 2(1 + \alpha^n + \beta^n + \gamma^n) \quad (6.6)$$

$$= 2\alpha^n + 2 + 4\alpha^{-n/2} \cos(n\theta). \quad (6.7)$$

In particular,

$$|\Phi_n - (2\alpha^n + 2)| \leq 4\alpha^{-n/2}, \quad \Phi_n \sim 2\alpha^n. \quad (6.8)$$

Proof. The power sum $\alpha^n + \beta^n + \gamma^n$ has initial values 3, 1, 3 by Newton's identities and obeys the Tribonacci recurrence. Hence it is L_n . Now apply Theorem 1.1. $\square$

The result is stronger than a conventional asymptotic expansion: it is an exact three-scale exponential representation. The subdominant pair produces a decaying oscillation rather than a monotone error.

Corollary 6.2 (Nearest-integer formula). *For every $n \geq 4$,*

$$\boxed{\frac{\Phi_n}{4} = \text{nint}\left(\frac{\alpha^n + 1}{2}\right)}, \quad (6.9)$$

where nint denotes the unique nearest integer.

Proof. By (6.7), the difference between the two sides before rounding is $\rho^n \cos(n\theta)$, whose absolute value is at most $\rho^n < 1/2$ for $n \geq 3$. $\square$

6.1 A convergent logarithmic transseries

Factoring the dominant term in (6.7) gives

$$\Phi_n = 2\alpha^n \left(1 + \alpha^{-n} + 2\alpha^{-3n/2} \cos(n\theta)\right). \quad (6.10)$$

Therefore, for all sufficiently large n (and in fact already in the stated range after direct checking),

$$\log \Phi_n = n \log \alpha + \log 2 + \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} \left(\alpha^{-n} + 2\alpha^{-3n/2} \cos(n\theta)\right)^m. \quad (6.11)$$

This is a convergent all-orders expansion, not merely a formal one. Its first layers are

$$\begin{aligned} \log \Phi_n = n \log \alpha + \log 2 + \alpha^{-n} + 2\alpha^{-3n/2} \cos(n\theta) - \frac{1}{2}\alpha^{-2n} \\ - 2\alpha^{-5n/2} \cos(n\theta) + \left(\frac{1}{3} - 2 \cos^2(n\theta)\right) \alpha^{-3n} + \mathcal{O}(\alpha^{-7n/2}). \end{aligned} \quad (6.12)$$

The half-integer exponential scales arise from the modulus $|\beta| = \alpha^{-1/2}$.

6.2 Asymptotics of the erroneous formula sequence

The composition numbers have the exact root expansion

$$F_n = \sum_{\lambda \in \{\alpha, \beta, \gamma\}} \frac{\lambda^{n+3}}{\lambda^2 + 2\lambda + 3}. \quad (6.13)$$

Substitution into (5.1) yields, for $n \geq 5$,

$$M_n = 1 + \sum_{\lambda \in \{\alpha, \beta, \gamma\}} \frac{\lambda^{n-1}(\lambda^2 + 2\lambda + 2)}{\lambda^2 + 2\lambda + 3}. \quad (6.14)$$

Thus

$$M_n \sim \frac{\alpha^2 + 2\alpha + 2}{\alpha(\alpha^2 + 2\alpha + 3)} \alpha^n = 0.4896527010215867240 \dots \alpha^n. \quad (6.15)$$

Since $R_n \sim \frac{1}{2}\alpha^n$, the limiting relative undercount is

$$1 - \lim_{n \rightarrow \infty} \frac{M_n}{R_n} = 0.0206945979568265519 \dots \quad (6.16)$$

So the discrepancy is small enough to be visually deceptive, but it does not vanish relatively.

7 Inverting the exact asymptotic transseries

The exact root formula permits a precise continuous inverse and, through it, a discrete threshold formula.

Define the real interpolation

$$\tilde{\Phi}(x) := 2 + 2\alpha^x + 4\alpha^{-x/2} \cos(\theta x), \quad x \geq 4. \quad (7.1)$$

It agrees with Φ_n at every integer $n \geq 4$.

[Write note, batch 7302, 1 October 2026.] Two remarks on this section. (1) Notation: the delivered text wrote L for $\log \alpha$ here, which collides with the Tribonacci–Lucas numbers L_n of Theorem 1.1; it is written $\Lambda = \log \alpha$ below. (2) Interpolation and provenance: $\tilde{\Phi}$ is one choice of real interpolation, and the continuous inverse $x(y)$ of Theorem 7.2 is the inverse of that choice. Between the values $y = \Phi_n$ its log-periodic coefficients $a_m(\psi)$ depend on the interpolation and are not determined by the sequence; what is intrinsic is the discrete inverse $N(y)$ of Corollary 7.3, which equals $\lceil x(y) \rceil$ for every strictly increasing interpolation. This is the dichotomy of Proposition `p9:prop:intrinsic` of the repository’s canonical transseries volume [11], stated there for the Fibonacci numbers. The mechanism itself, a dominant exponential with a complex-conjugate subdominant pair producing log-periodic inverse coefficients, is that volume’s Chapter p9 (Theorem `p9:thm:product` and Remark `p9:rem:shape`); no novelty is claimed for it here. The sequence-specific content is the explicit coefficients and the monotonicity of this interpolation.

Lemma 7.1 (Monotonicity of the interpolation). *The function $\tilde{\Phi}$ is strictly increasing on $[4, \infty)$.*

Proof. Put $\Lambda = \log \alpha$ and $B = \sqrt{(\Lambda/2)^2 + \theta^2}$. Then

$$\tilde{\Phi}'(x) \geq 2\Lambda\alpha^x - 4B\alpha^{-x/2}.$$

The polynomial in (6.1) is increasing for $x > 1$ and changes sign between 1.8 and 2, so $1.8 < \alpha < 2$. Hence $\Lambda > 0.58$, while $\theta < \pi$ and $B < \sqrt{(\log 2/2)^2 + \pi^2} < 3.17$. At $x = 4$ the displayed lower bound is therefore greater than

$$2(0.58)(1.8)^4 - 4(3.17)(1.8)^{-2} > 8.2.$$

The ratio of the possible negative term to the positive term decreases by a factor $\alpha^{-3/2}$ per unit increase of x , so the derivative remains positive. $\square$

For $y \geq 24$, let $x(y)$ be the unique real solution of $\tilde{\Phi}(x) = y$. Set

$$\begin{aligned} Y &:= \frac{y-2}{2}, & \Lambda &:= \log \alpha, \\ \omega &:= \frac{\theta}{\Lambda} = 3.5712382678640038805657341982470\dots, & \psi &:= \omega \log Y. \end{aligned} \tag{7.2}$$

Theorem 7.2 (Inverse log-periodic transseries). *As $y \rightarrow \infty$,*

$$\begin{aligned} x(y) &= \frac{\log Y}{\Lambda} - \frac{2 \cos \psi}{\Lambda} Y^{-3/2} \\ &\quad - \left(\frac{4 \cos^2 \psi}{\Lambda} + \frac{4\theta \cos \psi \sin \psi}{\Lambda^2} \right) Y^{-3} + \mathcal{O}(Y^{-9/2}). \end{aligned} \tag{7.3}$$

More generally, there is an all-orders expansion

$$x(y) = \frac{\log Y}{\Lambda} + \sum_{m \geq 1} a_m(\psi) Y^{-3m/2}, \tag{7.4}$$

where each a_m is a real-analytic 2π -periodic function computable recursively.

Proof. Write

$$x = \frac{\log Y}{\Lambda} + \delta, \quad t = Y^{-3/2}.$$

Dividing the equation $Y = \alpha^x + 2\alpha^{-x/2} \cos(\theta x)$ by Y gives

$$1 = e^{\Lambda\delta} + 2te^{-\Lambda\delta/2} \cos(\psi + \theta\delta). \quad (7.5)$$

At $t = 0$ the solution is $\delta = 0$, and the derivative of the right side with respect to δ is $\Lambda \neq 0$. The analytic implicit-function theorem therefore produces a convergent power series $\delta = \sum_{m \geq 1} a_m(\psi)t^m$ for small t , uniformly in the phase.

Substitute $\delta = a_1t + a_2t^2 + \mathcal{O}(t^3)$ into (7.5). Comparing the coefficient of t gives

$$\Lambda a_1 + 2 \cos \psi = 0.$$

Comparing the coefficient of t^2 then gives

$$\Lambda a_2 + 4 \cos^2 \psi + \frac{4\theta}{\Lambda} \cos \psi \sin \psi = 0.$$

These are exactly the two displayed correction terms. □

Corollary 7.3 (Discrete inverse). *For $y \geq 24$, define*

$$N(y) := \min\{n \geq 4 : \Phi_n \geq y\}.$$

Then

$$\boxed{N(y) = \lceil x(y) \rceil}. \quad (7.6)$$

Consequently, inserting (7.3) gives an asymptotically sharp threshold formula with a first correction of order $y^{-3/2}$ and a log-periodic phase.

Proof. Lemma 7.1 shows that $\tilde{\Phi}(n) \geq y$ if and only if $n \geq x(y)$. Since $\tilde{\Phi}(n) = \Phi_n$, the least such integer is $\lceil x(y) \rceil$. □

8 Independent exact verification

The supplied file `verify.py` (shipped as `code/verify.py`) performs an independent calculation using only the Python standard library. It does not use the recurrence proved in this article to compute the permanent.

For row i , a dynamic-programming state records:

- (a) the bit mask of already used target columns;
- (b) the exact integer sum $d_0 + \dots + d_i$.

Each transition chooses one of the four allowed targets $i, i+1, i+2, i+3 \pmod n$, provided that target has not been used. At the final row, the displacement sum is divided by n to recover the winding sector. All arithmetic is integral.

The computation through $n = 16$ returns:

n	$q = 0$	$q = 1$	$q = 2$	$q = 3$	total
4	1	11	11	1	24
5	1	21	21	1	44
6	1	39	39	1	80
7	1	71	71	1	144
8	1	131	131	1	264
9	1	241	241	1	484
10	1	443	443	1	888
11	1	815	815	1	1632
12	1	1499	1499	1	3000
13	1	2757	2757	1	5516
14	1	5071	5071	1	10144
15	1	9327	9327	1	18656
16	1	17155	17155	1	34312

The script also checks:

- both exact recurrences for the true count;
- the A000382 recurrence for the formula-defined sequence;
- both shifted generating-function numerators;
- the correction kernel coefficient by coefficient;
- the nearest-integer identity through $n = 100$ using an 80-digit Newton solve for α .

The captured output is included as `verification_output.txt` (shipped as `data/verification_output.txt`).

9 Proposed OEIS corrections

The following are mathematical recommendations, not submissions. Attribution, formatting, and historical priority should be reviewed by OEIS editors.

9.1 A000382

1. Remove the comment “The fourth column of A008305, divided by 4.” It is false beginning at $n = 8$.
2. Either rename the entry as the sequence defined by Mendelsohn’s displayed formula (4), divided by 4, or mark the historical provenance explicitly.
3. Replace “conjectured” on the affine recurrence by a proof reference to Theorem 5.1, provided the formula-defined meaning is adopted.
4. Label the rational function as the shifted tail generating function $\sum_{m \geq 0} a(m+4)z^m$.
5. Cross-reference the corrected quarter-count $R_n = (A001644(n) + 1)/2$ for $n \geq 4$ and A004306 for the undivided count.

9.2 A000496

1. Add that the terms from $n = 8$ onward are not the permanent $\text{per}(I + P + P^2 + P^3)$ and are not the fourth column of A008305.
2. Clarify that the displayed rational function is a tail generating function, not the ordinary generating function beginning at offset 0.
3. Retain the historical sequence if desired, but distinguish it from A004306.

9.3 A008305

In the cross-reference line for the fourth column, remove “4*A000382” and retain A004306. A direct comment may state that for $n \geq 4$

$$T(n, 4) = 2(A001644(n) + 1), \quad (9.1)$$

with sector sizes $(1, A001644(n), A001644(n), 1)$. *[Write note, batch 73O2: the identity (9.1) is already recorded in A004306 (“Equals 2*(A001644(n)+1), n>3”), so the proposal for A008305 amounts to cross-referencing the existing A004306 identity, with the sector sizes and a pointer to its proof.]*

9.4 A004306

The current entry already contains the correct recurrence and the equivalent Tribonacci formula. Useful additions would be:

$$\begin{aligned} A004306(n) &= 2(1 + \alpha^n + \beta^n + \gamma^n), \quad n \geq 4, \\ \frac{A004306(n)}{4} &= \text{nint}\left(\frac{\alpha^n + 1}{2}\right), \quad n \geq 4, \end{aligned}$$

and the winding-sector tiling interpretation from this report. *[Write note, batch 73O2: the entry’s cross-reference line already gives $2(A001644(n) + 1)$ for $n > 3$ and its formula line credits a tribonacci formula to Skirrow; the Binet and nearest-integer forms above restate that identity through $A001644(n) = \alpha^n + \beta^n + \gamma^n$, and the proposal is to cross-reference the existing identity and add the proof and the sector interpretation.]*

A plain-text draft suitable for editing is included as `oeis_corrections.txt` (shipped as `data/oeis_corrections.txt`; it predates these notes and does not mention the existing A004306 cross-reference).

10 Further research questions and topics

The winding decomposition is complete for $r = 4$ but exposes a larger program. The following problems are intended to be concrete enough for computation, proof, or formalization.

Research question 10.1 (Inner winding sectors for $r \geq 5$). For fixed r , determine the numbers

$$N_{n,r}(q) := \#\{\sigma : q(\sigma) = q\}$$

for $2 \leq q \leq r - 3$. The constant-cut law turns this into the enumeration of q -fold covers of a labeled cycle by intervals of lengths at most $r - 1$ with the additional bijectivity constraints at endpoints. Find a canonical local object—multitilings, cylindric paths, or a finite-state model—that makes those constraints transparent.

Research question 10.2 (Sector polynomial shape). The involution proves that $W_{n,r}(u)$ is palindromic. Is it unimodal or log-concave for all $n \geq r$? Is there a stability, real-rootedness, or γ -positivity phenomenon after a natural normalization? Small cases should be classified before making a universal conjecture.

Research question 10.3 (Minimal recurrences at fixed width). Classical transfer-matrix theory implies rational generating functions in n for each fixed r , but the minimal denominator and its factorization can be subtle. Determine the minimal recurrence for each individual sector and for the total $\Phi_{n,r}$, and explain cancellations bijectively.

Research question 10.4 (Dominant winding sector). For $r \geq 5$, which sector controls the exponential growth of $\Phi_{n,r}$? Does the maximizing sector concentrate near $(r - 1)/2$? Obtain the exponential rate and leading constant from a Perron–Frobenius operator, then compare with Mendelsohn’s general asymptotic claim.

Research question 10.5 (Weighted displacement enumerators). Study

$$\text{per}(w_0 I + w_1 P + \cdots + w_{r-1} P^{r-1})$$

sector by sector. For $r = 4$, the winding-one sector becomes a weighted cyclic composition polynomial. Determine whether the full four-sector formula has a useful multivariate factorization and whether it yields refined statistics on fixed points and jump lengths.

Research question 10.6 (Arbitrary displacement sets). Replace $\{0, 1, \dots, r - 1\}$ by an arbitrary subset $S \subset \mathbb{Z}/n\mathbb{Z}$. Characterize those S for which an extreme nonzero winding sector is a tiling language, and those for which the permanent admits a low-order recurrence independent of n .

Research question 10.7 (Other finite groups). Mendelsohn formulated displacements in abelian groups but specialized much of the computation to cyclic groups. Develop an analogue of the cut-flow argument on products of cycles, where cuts become codimension-one tori and the winding number becomes a homology class. Even the first nontrivial case $\mathbb{Z}_m \times \mathbb{Z}_n$ may reveal a useful higher-dimensional tiling model.

Research question 10.8 (A natural model for the historical “ghost” sequence). The A000382 tail has a perfectly valid positive recurrence and differs from the true quarter-count by cumulative Tribonacci numbers. Is there a natural class of cyclic tilings with a marked defect, boundary condition, or forbidden wrap that is counted by M_n ? A satisfactory model would preserve the historical sequence while preventing future confusion about its meaning.

Research question 10.9 (Divisibility without recurrence). The formula proves $4 \mid \Phi_n$. Find a direct free action or sign-reversing argument that partitions the allowed permutations into quadruples. Such an action might generalize to divisibility properties of $\Phi_{n,r}$.

Research question 10.10 (Inverse transseries for general rational recurrences). The inverse in Section 7 has periodic coefficients because the subdominant roots are a complex conjugate pair. Develop an automated theorem that takes a positive linear recurrence with a simple dominant

root and returns the inverse transseries, including resonances among subdominant root moduli and phases. *[Write note, batch 7302: partly re-scoped. Chapter p9 of the repository’s canonical transseries volume [11] already inverts a dominant exponential times finitely many exponentially small factors in closed form (Theorem p9:thm:product), explains the log-periodicity produced by a conjugate pair (Remark p9:rem:shape), and separates the forced part of the inverse from the interpolation-dependent part (Proposition p9:prop:intrinsic). What remains is the automated statement for an arbitrary positive linear recurrence, with its resonances.]*

Research question 10.11 (Formal verification). Formalize the constant-cut lemma, tiling bijection, and recurrence in Lean. The proof requires only finite sums, permutations of `Fin n`, modular arithmetic, and cyclic interval bookkeeping, making it a realistic addition to the ProveIt formalization program [10].

Research question 10.12 (Automated provenance audits). Build a reproducible tool that compares an OEIS definition against its b-file, formulae, cited tables, and independently computed terms. The present example shows why matching a long recurrence to inherited terms is not a substitute for recomputing the defining combinatorial object.

11 Conclusion

The four-displacement problem is governed by a simple topological statistic. The sum of clockwise displacements, divided by n , is an integer winding number, and local flow conservation forces every cut to have the same multiplicity. At winding one this turns the permutation into a cyclic tiling; for four allowed displacements, symmetry accounts for every remaining sector. The resulting exact count is

$$\Phi_n = 2(L_n + 1),$$

not the sequence currently attached to A000382 and A000496.

The historical error is unusually instructive. A published formula, a table in the same paper, and a later machine-guessed generating function all agree long enough to look persuasive, yet none of them replaces evaluation of the original permanent. Once the combinatorial structure is recovered, the correct recurrence, generating function, Binet formula, asymptotics, and inverse expansion all follow with little ambiguity.

A Proof details for the generating functions

For reference, multiplying a tail sequence $a_0, a_1, \dots$ by $1 - 2z + z^4$ produces coefficients

$$a_m - 2a_{m-1} + a_{m-4},$$

where missing terms are zero. For the true quarter-tail $6, 11, 20, 36, 66, \dots$, these coefficients are

$$6, -1, -2, -4, 0, 0, \dots,$$

which proves (4.11). For the A000382 tail $6, 11, 20, 36, 65, \dots$, they are

$$6, -1, -2, -4, -1, 0, 0, \dots,$$

which proves (5.3). Their difference is exactly z^4 over the common denominator.

B Reproduction instructions

The archive contains:

article.tex this source file;
article.pdf the compiled report;
verify.py independent exact enumeration and checks;
verification_output.txt
 captured verifier output;
oeis_corrections.txt
 editable proposed OEIS notes;
README.md a concise file guide and build command;
Makefile convenience targets for verification and compilation.

Run

```
python3 verify.py
pdflatex -interaction=nonstopmode article.tex
pdflatex -interaction=nonstopmode article.tex
```

or use the PDF build command documented in the README.

[*Write note, batch 7302, 1 October 2026.*] The list above describes the delivered archive. In ProveIt the report ships **article.tex**, **article.pdf**, a report **README.md** (replacing the delivered one), **code/verify.py**, **code/Makefile**, **data/verification_output.txt** and **data/oeis_corrections.txt**. The delivered README’s PDF build command and the **pdf** target of the **Makefile** call a script in the delivery environment that does not exist here; build with the two **pdflatex** commands above. Run the verifier as **python code/verify.py**; it writes only to standard output.

C Provenance

This report is one manuscript, number 55 of batch 7302 of ProveIt’s incoming-reports intake: the archive **ProveIt_A000382_winding_correction.zip** (19-page PDF), dated October 1, 2026, which arrived in commit **58dbe2cd3** and was placed by commit **6e193dd4f**. It cites no repository revision. The text is as delivered except for the label prefix **wdc:**, the renaming of $\log \alpha$ from L to Λ in Section 7, the citation of the repository (listed but uncited in the delivered bibliography), the shipped file names, this appendix, one bibliography entry, and the dated write notes, which record that A004306 already prints $\Phi_n = 2(A001644(n) + 1)$, that Mendelsohn’s values are read from a scan, and the interpolation dependence and repository provenance of the inverse.

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- [11] ProveIt, *Transseries: the polynomial-logarithmic calculus, series reversal at infinity, and inversion*, Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/, Chapter p9 (*A real-argument Fibonacci function*). Cited in the write (batch 73O2), not by the manuscript.